

2 (a) (i) Find resistance

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P} = 6\Omega$$

(ii) Energy transferred

$$E = Pt$$

$$E = (24)(1800) = 4.32 \times 10^4 \text{ J} = 4.3 \times 10^4 \text{ J}$$

(iii) The total amount of charge through lamp

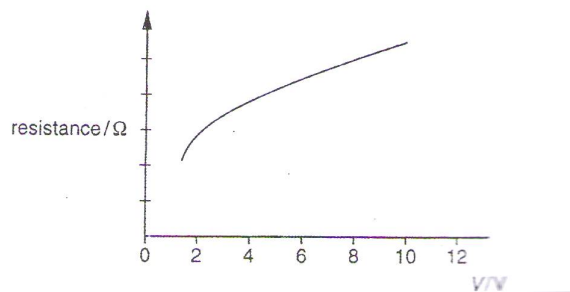
$$Q = \frac{E}{V}$$

$$Q = \frac{4.32 \times 10^4 \text{ J}}{12} = 3600 \text{ C}$$

$$\text{Hence the number of electrons} = \frac{3600}{1.6 \times 10^{-19}} = 2.25 \times 10^{22} = 2.3 \times 10^{22}$$

(b) (i) The resistance (R) at any particular set of values of pd (V) and current (I) is the ratio of the potential difference to current, $R = V/I$.

(ii)



Note: Many ignored instruction and calculated resistance values. Majority gave graphs starting from origin or negative intercept!

(iii) The pd across the lamp cannot be 0 V even with the variable resistor set to its maximum value (10Ω), as the circuit resistance is not infinite and there will still be current through the lamp.

The pd across the lamp cannot be 12 V even with the variable resistor set to its minimum value (0Ω) because there could be pd across the internal resistance of the cell and the ammeter, which may not be ideal.